

## **Lateral migration and segregation of vesicles with viscosity contrast in simple shear and Poiseuille flows**

Gwennou Coupier<sup>1</sup>, Natacha Callens<sup>2</sup>, Badr Kaoui<sup>1,3</sup>, Christophe Minetti<sup>2</sup>, Frank Dubois<sup>2</sup>, Chaouqi Misbah<sup>1</sup> and Thomas Podgorski<sup>1</sup>

<sup>1</sup>Laboratoire de Spectrométrie Physique, Université Joseph Fourier (Grenoble I) and CNRS, 140 Rue de la Physique, F-38402 St Martin d'Hères Cedex, France

<sup>2</sup>Microgravity Research Center, Université Libre de Bruxelles, 50 Av. F. Roosevelt, CP 165/62, B-1050 Brussels, Belgium

<sup>3</sup>Université Hassan II - Mohammedia, Faculté des Sciences Ben M'Sik, Laboratoire de Physique de la Matière Condensée, BP 7955 Casablanca, Morocco

### **ABSTRACT**

Lateral migration of vesicles (closed lipidic membranes) in a flow is characterized as a function of the relevant flow parameters and mechanical properties of the vesicles. We consider low Reynolds number flows, and migration is only due to viscous effects. Through experiments and simulations, we exhibit two different origins for such cross streamline migration: the presence of a wall, and a non-constant shear rate, as for instance in a Poiseuille flow. Such migration modifies the distribution of vesicles in a sheared polydisperse suspension; we present preliminary results proving that the hydrodynamic interactions between vesicles greatly modify the distribution of vesicles according to their sizes and deflation.

### **INTRODUCTION**

The ability of soft entities such as vesicles, drops, capsules or blood cells, to adapt their shapes under non-equilibrium conditions gives them the possibility to migrate transversally to the streamlines even at low Reynolds number. Those transverse migrations induce non-uniform lateral distributions of the suspended objects, which has important consequences on the rheology of a confined suspension, as illustrated for instance the Fahraeus-Lindquist effect in blood vessels [1].

Vesicles are closed phospholipid membranes; they encapsulate an internal fluid and are usually suspended in an external aqueous solution. The physics of vesicles has attracted much interest in the recent decades and have been the subject of many theoretical and experimental works. This is because of their ability to reproduce some dynamical behaviors observed for living cells (such as red blood cells [2]) and the interest to exploit them as carriers of biomaterials. Vesicle dynamics in shear flow has been the subject of extensive studies [3-9]. A vesicle placed in an unbounded fluid subject to simple shear does not exhibit any lateral migration with respect to the flow direction. When the viscosity ratio between the inner and the outer fluids is small, it performs a tank-treading dynamics where the orientation of the main axis of the vesicle is constant and the membrane undergoes a tank-treading motion.

We shall see in the following that once this flow is bounded by a wall and/or the velocity profile is not linear (non homogeneous shear rate), tank-treading vesicles experiment transverse forces inducing lateral migration, even in the Stokes limit, when inertial effects can be neglected. We present a brief summary of our recent experiments on isolated vesicles migrating in a

bounded shear flow and in a channel (bounded Poiseuille flow). We then discuss the effects of interactions between vesicles on their lateral distribution in a simple shear flow chamber.

## EXPERIMENTS

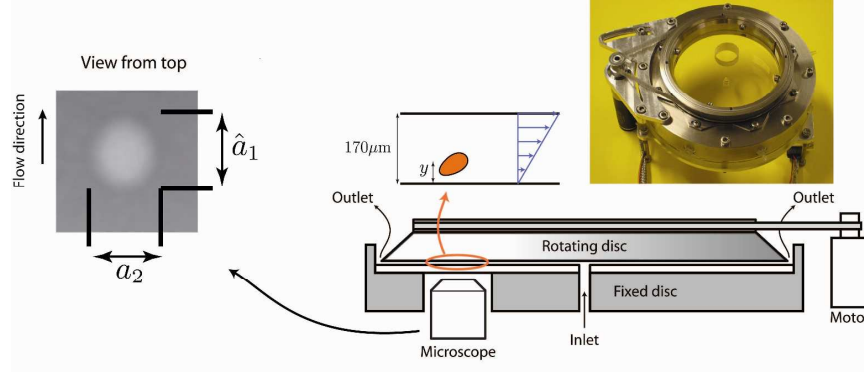
A vesicle, whose membrane cannot be stretched, is characterized by two geometrical parameters: its effective radius  $R_0$ , determined from its incompressible volume  $V$  by  $R_0 = (3V/4\pi)^{1/3}$  and its reduced volume  $\nu = V/[4\pi(S/4\pi)^{3/2}/3]$  ( $S$  is the area of the vesicle), characterizing vesicle deflation. The reduced volume is the vesicle volume divided by the volume of a sphere of same area. The viscosity ratio is defined as  $\lambda = \eta_{in}/\eta_{out}$ , where  $\eta_{in}$  and  $\eta_{out}$  denote the inner and the outer viscosity, respectively. Vesicles are prepared following the electroformation method [10]. They are made of a dioleoylphosphatidylcholine (DOPC) lipid bilayer enclosing an inner solution of sugar (sucrose or glucose) in water or in a 1:4 glycerol-water (w:w) mixture. Samples are diluted in a slightly hyperosmotic outer solution of the same type, in order to deflate them by osmosis. Dextran can be added to one of the solutions to modify the viscosity ratio  $\lambda$ .  $R_0$  is typically between 5 and 40 microns, while the viscosities lie between 1 and 20 mPa.s. Thanks to the possibility to modify easily the solutions to control the deflation and the viscosity contrast, vesicles are convenient controllable tools to study the generic behavior of soft deformable entities in a flow. However, it is often impossible to make the vesicles buoyant at the same time. If one wishes to study the hydrodynamic interactions with no screening by gravity, one has to resort to study quasi-2D movements (with negligible motion in the gravity direction), or to work under microgravity conditions.

Vesicles behavior in Poiseuille flow was studied in a horizontal straight microfluidic channel of rectangular cross section (each dimension being around 100 microns). The walls of the channels are made of PDMS glued to a glass cover slide. The flow is induced by gravity, by connecting the inlet and the outlet to reservoirs at different heights. Vesicles are tracked along their trajectory thanks to a phase contrast microscope. Since, as we shall see, the migration movements results in a centering of the vesicles, we waited for the flow to have been established for long, resulting in quasi-centering of vesicles in the direction of gravity. In such confined situation hydrodynamics forces generally dominate over the weight. On the contrary, a particular design of the upstream channel created an initial condition where incoming vesicles were not already centered in the other cross direction, thus allowing the study of their migration in a quasi-2D situation [11].

The study of vesicles under bounded simple shear flow in microgravity conditions is the goal of ESA project BIOMICS (BIOMImetic and Cellular Systems). Within this framework several parabolic flight campaigns on board the Airbus A300-Zero G from Novespace (Mérignac, France) were performed. The 22s-long microgravity periods allowed us to study the lift of isolated vesicles from a bottom wall [12]. However, this timespan is not sufficient to study stationary states where more concentrated vesicles have reached their final lateral position between the two walls of the shear flow chamber, but still feel the influence of their neighbors. This situation was studied by participating to the MASER 11 sounding rocket flight in May 2008, which offered us a continuous microgravity period of nearly 6 minutes.

The shear flow chamber was designed and manufactured by the Swedish Space Corporation. As shown on figure 1, it is made of two parallel glass discs, with a diameter of 10 cm and a gap of 170  $\mu\text{m}$ . The bottom disc is fixed and the top one can be rotated at constant

speed, allowing shear rates between  $0.5$  and  $50 \text{ s}^{-1}$ . An inlet at the center of the bottom plate and several outlets at the periphery allow filling with vesicles suspensions and rinsing. A fixed digital holographic microscope (DHM), working in transmission with a partial spatial coherent source, allowed a 3D visualization of the evolution of vesicle positions in the field of view during microgravity [13]. Note that in such a configuration, we see a projection of the vesicles on the plane perpendicular to the optical axis and to migration direction. Thus, the deflation of the vesicles are measured by  $a_2 / \hat{a}_1$ , where  $\hat{a}_1$  is the apparent big axis of the vesicle, and  $a_2$  one of its small axis (figure 1). In all the experiments, we consider the  $y$  position of the center of mass of the vesicles relative to one of the two bounding walls.



**Figure 1.** Experimental set-up allowing the study of single vesicles or suspensions in a bounded simple shear flow. The use of digital holographic microscopy allows to reconstruct the  $y$  positions of the vesicles.

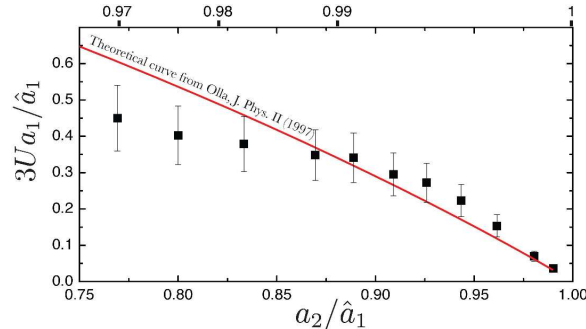
## DISCUSSION

### One vesicle in a simple shear flow

A vesicle placed in a bounded simple shear flow experiments a lift force that pushes it far from the walls. This viscous lift force is caused by the flow induced fore-aft symmetry breaking of the vesicle's shape [14]. We showed that for  $\lambda = 1$  the lift velocity due to one wall is well described by the following law, in agreement with refs. [15,16]:

$$\dot{y} = U(\lambda, \nu) \dot{\gamma} \frac{R_0^3}{y^2}, \quad (1)$$

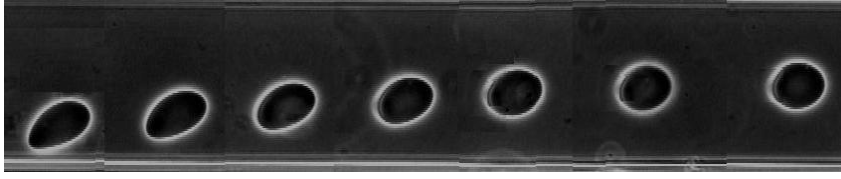
where  $\dot{\gamma}$  is the flow shear rate and  $U$  is the dimensionless lift velocity [12]. Thus, the vesicle will eventually reach a stationary position at equal distance from the two walls.  $U$  was shown to increase with decreasing  $\nu$ : higher deflation leads to higher capability to migrate (figure 2). Investigations on the  $\lambda$  dependency of  $U$  are planned.



**Figure 2.** Variations of the lift velocity with the deflation of the vesicle. While a spherical vesicle does not experiment any lift force due to its fore-aft symmetry, migration is enhanced by deflation. Experimental points are in good agreement (no fitting parameter) with the theoretical values proposed in [15] (full line).

## One vesicle in a Poiseuille flow

As shown on figure 3, a tank-treading vesicle placed in a bounded Poiseuille flow migrates towards the centerline of the channel. Indeed, this behavior is due to two contributions: in addition to the wall induced lift, the non linear character of the Poiseuille flow velocity profile (non uniform shear rate) induces lateral migration of vesicles towards the flow centerline, even in unbounded geometry, as showed our boundary integral method simulations [17]. In such unbounded case, the migration velocity is constant along the trajectory, except near the centerline where it vanishes.

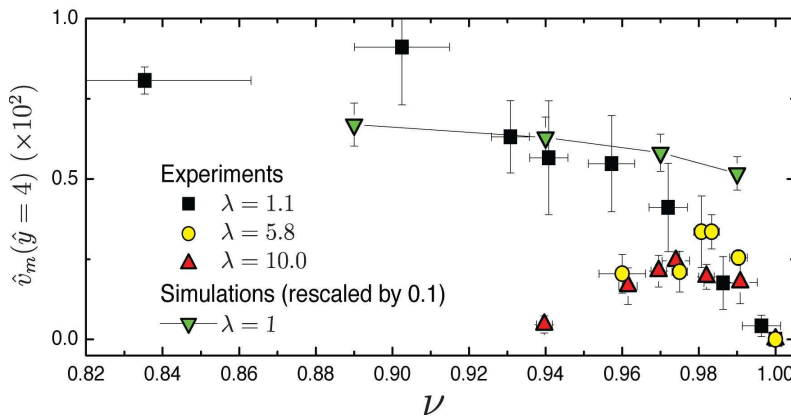


**Figure 3.** Migrating vesicle in a channel of width 140  $\mu\text{m}$  ( $\lambda = 1$ ).

The coupling between these two effects is non-trivial, since the presence of walls and the variations in the shear rate induce continuous shape changes, which affect the forces on the vesicle. This results in a different migrating law, given by:

$$\dot{y} = U(\lambda, \nu) \dot{\gamma}(y) \frac{R_0^{\delta+1}}{(y - R_0)^\delta}, \quad (2)$$

where for  $\lambda = 1$ ,  $\delta = 1 \pm 0.1$  [11]. Exhaustive study on the  $(\lambda, \nu)$  dependency of  $\delta$  and  $U$  is under way. Nevertheless, it is noteworthy that the exponent  $\delta$  is markedly different from 2, as found in a simple shear flow. If we consider the migration velocity at a given point for a vesicle of given size (figure 4), we found also that for  $\lambda \leq 2$ , deflating the vesicle increases its velocity. However, for higher  $\lambda$ , we found a non monotonous behavior of the migration velocity with  $\nu$ . Deflating too much leads to a decrease of the velocity. This is due to the fact that the main axis of more viscous vesicles has a smaller angle relatively to the flow direction, in order to minimize viscous dissipation. It approaches the 0 value, which is a situation where the fore-aft symmetry is recovered, and where no migration is expected, as for spherical objects ( $\nu = 1$ ).

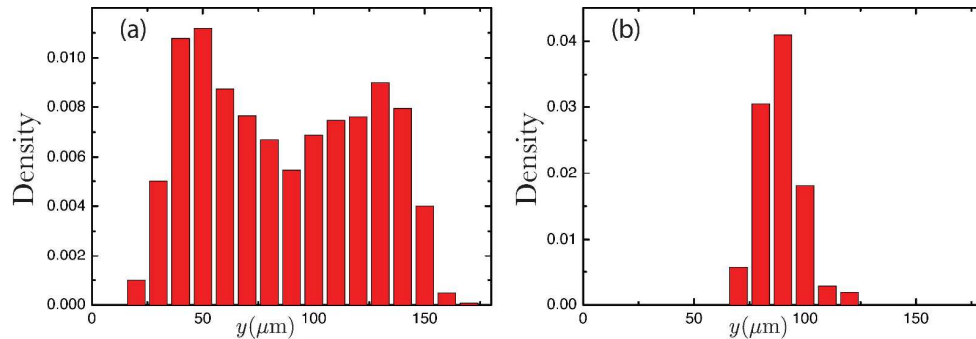


**Figure 4.** Variations with  $\nu$  of the reduced velocity  $\hat{v}_m = \dot{y}/[R_0 \dot{\gamma}(y)]$  at position  $\hat{y} = y/R_0 = 4$  for different viscosity ratios. Simulation points are available for  $\lambda = 1$  but the absolute values are higher due to the 2D character of our model [11].

## Equilibrium distribution in a vesicles suspension under shear

The distribution of vesicles in a suspension sheared between plates was investigated in microgravity. After some time, an equilibrium is reached between the lift forces that push vesicles away from walls, and hydrodynamic interactions between vesicles that lead to shear induced diffusion, which increases with vesicles size [18, 19].

As seen on figure 5, the influence of vesicle size and shape on the distribution of objects in the thickness of the flow chamber is very strong: small and nearly spherical vesicles have wider distributions than big deflated ones which are concentrated near the middle plane. This is coherent with the dependence of the lift velocity, which is in balance with repulsion, on these parameters (eq. 1).



**Figure 5.** Distribution of vesicles across the shear flow chamber thickness (distance between discs: 170  $\mu\text{m}$ ), for a viscosity ratio of 3.9, a shear rate of  $50 \text{ s}^{-1}$ , and a volumic fraction of 0.03% of vesicles. (a): small, nearly spherical vesicles ( $1 < R_0 < 4 \mu\text{m}$ ,  $0.91 < a_2 / \hat{a}_1 < 1$ ), (b): bigger, deflated vesicles ( $5 < R_0 < 10 \mu\text{m}$ ,  $0.67 < a_2 / \hat{a}_1 < 0.83$ ).

A segregation effect is also observed in these polydisperse suspensions. As seen in figure 5, small vesicles exhibit a bimodal distribution with two maxima of density and a minimum at the center while the distribution of big vesicles is a narrow, centered peak. While small vesicles migrate at a much smaller speed towards the center, they thus also undergo a strong repulsion from hydrodynamic interactions with big vesicles and are expelled from the center. It is noteworthy that this segregation effect is clear while the volumic fraction in vesicles is as low as 0.03%.

This segregation could be similar to that observed in some blood flows between red blood cells and platelets. Further analysis of the results should provide more quantitative information on this phenomenon.

## CONCLUSIONS

We described two different mechanisms for vesicle migration in a flow, namely wall-induced lift and centering in a parabolic velocity profile. In a channel, both effects contribute to center the vesicle, but the coupling between them is nonlinear. We proposed scaling laws for the lift velocity as a function of the shear rate, vesicle's size and position. The parameters of these laws depend on the vesicle's viscosity contrast and deflation. In particular, while deflating a vesicle first increases its ability to migrate quickly, more deflated vesicles whose inner viscosity exceeds a threshold do not migrate as efficiently. Deflation and relative sizes are key parameters

for the hydrodynamic interactions between vesicles: in a sheared suspension, small and quasi-spherical vesicles are expelled from the center by larger or more deflated one. More quantified analysis of this effect should provide a basis for discussion of segregation in a polydisperse suspension of deformable objects. Investigation on the rheological signature of such phenomenon is the natural continuation of this work.

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